

LLM-Based Extraction of Pharmacogenomic Gene-Drug Associations from PharmGKB Text: A Feature-Fusion Approach for Warfarin Dose Prediction

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Abstract

Structured pharmacogenomic decision-support models typically rely on curated genotype, demographic, and clinical features, leaving a substantial amount of evidence embedded in unstructured text — such as PharmGKB annotations and biomedical literature — unused. We present a two-stage pipeline in which Qwen2.5-3B-Instruct, fine-tuned with LoRA, is used as an evidence-extraction layer to classify gene-drug association status directly from PharmGKB text passages, and the resulting extraction features are combined with structured clinical/genotype features to support a downstream warfarin dose-prediction task using IWPC data, with XGBoost performing the downstream prediction. On a held-out test set of 1,655 gene-drug pairs, 216 pairs had ambiguous gold association labels and were excluded from binary evaluation, leaving 1,439 unambiguous instances on which the extraction model achieves 84.09% accuracy ($F1 = 85.02\%$). Applied to warfarin dose prediction on 658 records from 333 subjects (subject-level held-out split), the fused model achieves $R^2 = 0.70$, $MAE = 7.0$ mg/week, and $RMSE = 10.0$ mg/week. We report these results alongside their limitations, including the absence of a structured-only ablation baseline, and position this work relative to prior pharmacogenomic text-mining and LLM-based extraction efforts rather than claiming feature fusion itself as novel.

1 Introduction

Pharmacogenomic decision support — using a patient’s genetic profile to guide drug selection and dosing — depends on evidence linking specific genes to specific drug responses. Databases such as PharmGKB curate this evidence, but curation is manual, and much of the underlying evidence exists only as unstructured text: annotation sentences, literature excerpts, and clinical notes. Structured predictive models built on top of these databases (e.g., XGBoost-based recommenders using genotype and clinical features) therefore do not directly use the textual evidence itself, only its curated, structured summary.

This report investigates whether a fine-tuned large language model, used strictly as an evidence-extraction tool, can extract gene-drug association signal directly from PharmGKB text, and whether that extracted signal, when combined with existing structured features, is useful for a downstream clinical prediction task. We use warfarin dose prediction as the downstream application, since it is a well-studied pharmacogenomic dosing problem with an available public dataset (IWPC). Qwen is used exclusively as an evidence-extraction model in this pipeline; it does not make clinical decisions, recommend treatments, or predict dose directly. That role is performed by the downstream XGBoost model.

The primary contribution of this work is not a new feature-fusion architecture itself, but the application of an LLM-based pharmacogenomic evidence extraction pipeline to a downstream quantitative pharmacogenomic prediction task. Concretely:

1. Qwen2.5-3B-Instruct is fine-tuned with LoRA to extract structured association information from PharmGKB evidence sentences, evaluated on a pair-aware held-out split.

2. The extracted evidence is converted into numerical features that can be combined with genotype-derived variables.
3. These combined features are used with XGBoost for downstream warfarin dose prediction, evaluated on a subject-level held-out split.
4. We report an honest accounting of what remains unvalidated — most notably, the lack of a structured-only ablation to isolate the extraction features’ contribution.

The contribution is therefore the specific connection between curated PharmGKB evidence, LLM-based structured extraction, genotype information, and downstream warfarin dose prediction, rather than the invention of feature fusion as a technique.

2 Related Work

2.1 Pharmacogenomic Knowledge Extraction

Automated extraction of pharmacogenomic evidence from text is an active area. PGxMine [1] is prior work specifically targeting automated extraction pipelines for PharmGKB-style annotations, aiming to identify pharmacogenomic entities and relationships from biomedical literature at scale, focused primarily on entity-level extraction rather than downstream prediction.

2.2 PharmGKB Evidence and Structured Pharmacogenomic Knowledge

PharmGKB provides curated pharmacogenomic evidence and gene-drug relationships, but individual annotation sentences typically contain considerably more detail than a curated relationship label alone — including the specific genetic variant involved, phenotype, direction of effect, and whether the relationship concerns a pharmacokinetic (PK), pharmacodynamic (PD), efficacy, toxicity, metabolism, or other phenotype/effect relationship. Extracting structured information directly from these evidence sentences therefore has the potential to provide a richer representation than a simple binary gene-drug association label, though realizing that richer representation is not the task undertaken in this report (see Section 4).

2.3 Warfarin Dose Prediction

The International Warfarin Pharmacogenetics Consortium (IWPC) established the clinical importance of pharmacogenetic variables — most notably *CYP2C9* and *VKORC1* genotype — for predicting individualized warfarin dose, alongside clinical and demographic variables, via a pharmacogenetic dosing algorithm validated against fixed-dose and clinical-only approaches [2]. Subsequent work has applied a range of machine learning and tree-based approaches, including gradient-boosted methods such as XGBoost, to warfarin dose prediction using IWPC and related cohorts.

2.4 LLM and Structured-Feature Fusion

Recent work has explored LLM-based, evidence-grounded systems for pharmacogenomic recommendation generation directly from structured knowledge bases and free-text literature, including agentic pipelines that synthesize guideline-style recommendations rather than stopping at extraction [4]. Separately, combining LLM-derived or text-derived signal with structured numerical features has precedent in adjacent biomedical prediction tasks: LLM-DDI combines LLM-derived embeddings with a message-passing graph neural network to improve drug-drug interaction prediction on a biomedical knowledge graph [3], and earlier work has used BioBERT-based extraction fused with structured features for related biomedical prediction tasks. Feature fusion of text-derived and structured signal is therefore not itself a novel technique, and is not claimed as one in this report.

2.5 Positioning of the Present Study

Prior literature provides foundations for pharmacogenomic text mining, PharmGKB evidence curation, LLM-based biomedical extraction, IWPC-based warfarin prediction, and machine-learning prediction using structured variables. The present study connects these components into a specific workflow in which Qwen acts as an evidence-extraction layer and XGBoost acts as the downstream predictive model. The key contribution is therefore the application of LLM-based PharmGKB evidence extraction to augment structured genotype information for downstream warfarin dose prediction, rather than the invention of feature fusion itself.

3 Data

Extraction dataset. Text passages are drawn from PharmGKB gene-drug relationship annotations, covering 11,035 gene-drug pairs. The dataset is split pair-aware into training (7,725 pairs), validation (1,655 pairs), and test (1,655 pairs) sets, with sentence-level annotations assigned according to their parent gene-drug pair. This prevents any gene-drug pair from appearing in both training and test data, avoiding inflated performance from pair memorization.

Downstream dosing dataset. For the warfarin dose-prediction application, we use IWPC data comprising 658 records from 333 unique subjects. The train/test split is performed at the subject level, ensuring no patient’s records appear in both sets.

4 Method

4.1 Binary association task definition

Qwen is used strictly as an evidence-extraction model, not as a clinical decision-maker or drug recommendation system. The extraction task evaluated in this report is a coarse binary classification of association status, defined as follows: an *associated* instance is a gene-drug pair whose PharmGKB-derived gold relation is explicitly labeled as associated; a *not associated* instance is a gene-drug pair whose PharmGKB-derived gold relation is explicitly labeled as not associated. Instances with ambiguous gold relations are excluded from this binary evaluation (see Section 5.1 for counts).

This binary association-status task does not distinguish biological relation types such as pharmacokinetic (PK), pharmacodynamic (PD), efficacy, toxicity, metabolism, or other phenotype/effect relationships, and should not be interpreted as full pharmacogenomic relation extraction. It is a coarser task than relation-type classification, and we treat it as such throughout this report.

4.2 Extraction pipeline

We fine-tune Qwen2.5-3B-Instruct using LoRA (PEFT) to classify gene-drug association status (associated / not associated / ambiguous, with ambiguous instances excluded from binary evaluation) from PharmGKB text passages. Training configuration is summarized in Table 1.

4.3 Baseline recommender

As prior work, we use a structured XGBoost-based drug recommendation model trained on genotype, demographic, and clinical features (Sakhalkar et al., prior work).

4.4 Integration

Qwen-derived association predictions are converted into engineered features (e.g., association presence, model confidence) and concatenated with structured clinical/genotype features. This fused feature set is

Table 1: LoRA fine-tuning configuration for the extraction model.

Hyperparameter	Value
LoRA rank (r)	16
LoRA alpha	32
LoRA dropout	0.05
Target modules	q_proj, k_proj, v_proj, o_proj
Quantization	4-bit NF4
Learning rate	1×10^{-4}
Training epochs	2
Per-device batch size	2
Gradient accumulation steps	8

used to train an XGBoost model for the downstream warfarin dose-prediction task. Qwen performs evidence extraction only; XGBoost performs the downstream prediction.

5 Results

5.1 Extraction / association classification performance

The held-out test set contained 1,655 gene-drug pairs. Of these, 216 had ambiguous gold association labels and were excluded from the binary associated/not-associated evaluation, leaving 1,439 unambiguous instances for calculation of the confusion matrix and classification metrics reported below. On this 1,439-instance unambiguous binary evaluation subset, the fine-tuned Qwen2.5-3B-Instruct model achieved the results in Table 2.

Table 2: Qwen binary association-status classification performance on the 1,439-instance unambiguous evaluation subset.

Metric	Value
Accuracy	84.09%
Precision	83.44%
Recall	86.67%
F1	85.02%

Table 3: Confusion matrix on the 1,439-instance unambiguous evaluation subset.

	Predicted Negative	Predicted Positive
Actual Negative	560 (TN)	129 (FP)
Actual Positive	100 (FN)	650 (TP)

5.2 Downstream application: warfarin dose prediction

To assess whether extracted pharmacogenomic evidence is useful for a downstream quantitative task, we apply the fused feature pipeline (structured features + Qwen-derived association features) to warfarin therapeutic dose prediction using IWPC data (658 records, 333 subjects, subject-level held-out split). The fused

model achieves the results in Table 4.

Table 4: Fused model performance on warfarin dose prediction.

Metric	Value
R^2	0.70
MAE	7.0 mg/week
RMSE	10.0 mg/week

We did not run a structured-features-only baseline for this task, so we cannot directly quantify the contribution of the Qwen-derived features to this specific result. For external context, published IWPC pharmacogenetic dosing algorithms have reported R^2 in a comparable range on similar cohorts; a direct ablation against a structured-only baseline is left as future work. This $R^2 = 0.70$ figure describes the downstream XGBoost regression model’s performance and should not be read as an accuracy figure for Qwen’s association classification, which is reported separately in Section 5.1.

6 Discussion and Limitations

The Qwen association-classification result (84.09% accuracy on the unambiguous binary evaluation subset) should not be described as a clinical accuracy figure; it reflects performance on a coarse, sentence-level, binary evidence-extraction task, not a clinical outcome prediction. It also should not be characterized as full pharmacogenomic relation extraction, since it does not distinguish PK, PD, efficacy, toxicity, metabolism, or other relation types.

The warfarin dosing evaluation, while a real and encouraging result, is limited by sample size (658 records, 333 subjects) relative to the scale typically used in dosing-algorithm literature, which constrains how confidently the result generalizes to broader patient populations. Without a structured-only ablation on the same dosing task, we cannot yet isolate how much of the $R^2 = 0.70$ result is attributable to the Qwen-derived features specifically versus the structured features alone.

This work has not been clinically validated and is not intended for clinical use. Qwen does not recommend drugs or predict warfarin dose directly; it extracts association evidence that is one input among several to a separate downstream predictive model. As with any pharmacogenomic decision-support tool, errors in extracted evidence — particularly false positives on gene-drug associations — carry real downstream risk if such a system were ever used to inform actual dosing decisions.

7 Future Work

Future work includes: (1) evaluating the model on larger and more diverse pharmacogenomic datasets to improve generalizability across populations and drug classes; (2) running a structured-only ablation on the warfarin dosing task to isolate the contribution of Qwen-derived features; (3) extending extraction beyond binary association classification to full relation typing (PK/PD, efficacy, toxicity, metabolism, and other effect categories); and (4) investigating larger models or alternative parameter-efficient fine-tuning strategies to improve extraction of complex, conditional, or negated genotype-drug relationships, as well as to reduce the proportion of instances requiring exclusion as ambiguous.

8 Conclusion

We present a pipeline in which Qwen2.5-3B-Instruct is fine-tuned as an evidence-extraction layer to classify gene-drug association status from PharmGKB text, with the resulting features combined with structured

genotype and clinical variables for downstream warfarin dose prediction using XGBoost. On the 1,439-instance unambiguous binary evaluation subset (of 1,655 held-out gene-drug pairs, 216 excluded as ambiguous), the extraction model achieves 84.09% accuracy. The fused downstream model achieves $R^2 = 0.70$ on a subject-level held-out warfarin dosing evaluation. We report these results alongside their limitations, most notably the absence of a structured-only ablation, and position this work within — rather than as an extension of the novelty of — the broader, active line of research on LLM-based pharmacogenomic evidence extraction and structured-feature fusion.

References

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